AC Circuit Part 2

Loads of AC Circuits

1. Purely resistive loads.

Important properties of purely resistive loads:

a) The current is always **in phase** with the voltage (phase angle is zero).
b) The current I = V/R where V and I are rms values.
c) The power consumed (dissipated) = VI= I^2R = V^2/R.

2. Purely inductive loads.

Important properties of purely inductive loads:

a) The current always lags behind voltage by exactly 90 electrical degrees (ELI).
b) The current $I = \frac{V}{X_L} = \frac{V}{2\pi f L}$.
c) The power consumed (dissipated) is zero.

3. Purely capacitive loads.

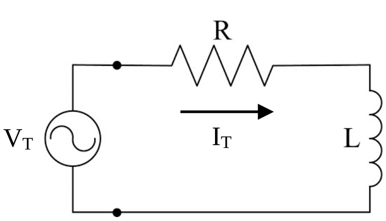
Important properties of purely capacitive loads:

a) The current always leads the voltage by exactly 90 electrical degrees (ICE).
b) The current $I = \frac{V}{X_C} = (V)(2\pi f C)$.
c) The power consumed (dissipated) is zero.

Series AC Circuits

When dealing with series AC circuits, use the current phasor as the reference phasor since the same current flows through all the components. The phasor sum of the various voltage drops is equal to the applied voltage.

Series RL Circuit



Formulas:

Total voltage, V_T

$$V_T = V_R + jV_L = |V_T|\angle\theta$$
$$|V_T| = \sqrt{V_R^2 + V_L^2}$$
$$\theta = \tan^{-1}\frac{V_L}{V_R}$$

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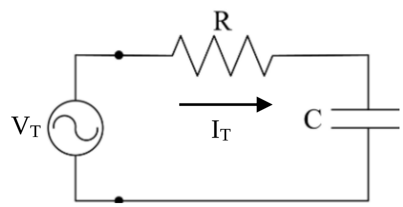
Total impedance, Z

$$Z = R + jX_L = |Z|\angle\theta$$
$$|Z| = \sqrt{R^2 + X_L^2}$$
$$\theta = \tan^{-1}\frac{X_L}{R}$$

Total current, I_T $\quad I_T = \frac{V_T}{Z}$

Series RC Circuit



Formulas:

Total voltage, V_T

$$V_T = V_R - jV_C = |V_T|\angle\theta$$
$$|V_T| = \sqrt{V_R^2 + V_C^2}$$
$$\theta = -\tan^{-1}\frac{V_C}{V_R}$$

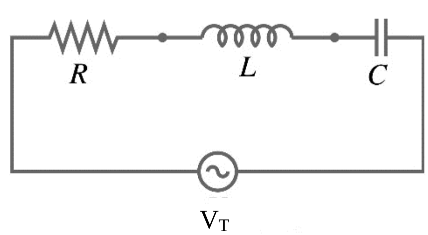
Note:
-θ indicates that the voltage is lagging behind the current.

Total impedance, Z

$$Z = R - jX_C = |Z|\angle\theta$$
$$|Z| = \sqrt{R^2 + X_C^2}$$
$$\theta = -\tan^{-1}\frac{X_C}{R}$$

Total current, I_T $\quad I_T = \frac{V_T}{Z}$

Series RLC circuit



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Total voltage, V_T

$$V_T = V_R + jV_L - jV_C = |V_T|\angle\theta$$
$$|V_T| = \sqrt{V_R^2 + (V_L - V_C)^2}$$
$$\theta = \pm\tan^{-1}\frac{(V_L - V_C)}{V_R}$$

Total impedance, Z

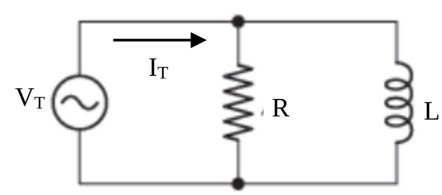
$$Z = R + jX_L - jX_C = |Z|\angle\theta$$
$$|Z| = \sqrt{R^2 + (X_L - X_C)^2}$$
$$\theta = \pm\tan^{-1}\frac{(X_L - X_C)}{R}$$

Total current, I_T $\quad I_T = \frac{V_T}{Z}$

Parallel AC Circuits

When dealing with parallel AC circuits, use the voltage source as the reference phasor since the same voltage appears across each component. The total current is equal to the phasor sum of branch currents.

Parallel RL Circuit



Total current, I_T

$$I_T = I_R - jI_L = |I_T|\angle\theta$$
$$|I_T| = \sqrt{I_R^2 + I_L^2}$$
$$\theta = -\tan^{-1}\frac{I_L}{I_R}$$

Note:
-θ indicates that current I_T is lagging behind the voltage V_T.

Total admittance, Y

$$Y = G - jB_L = |Y|\angle\theta$$
$$|Y| = \sqrt{G^2 + B_L^2}$$
$$\theta = -\tan^{-1}\frac{B_L}{G}$$

Total impedance, Z $\quad Z = \frac{1}{Y}$

Total voltage, V_T $\quad V_T = I_T Z$